

Felicity tech tree: low temperature electrolysis of mixed metal oxides

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Abstract

Math behind the process for extracting metal from asteroid regolith with low temperature electrolysis.

Largely based on my interpretation of the paper "[Low Temperature Electrolysis of Mixed Metal Oxides](#)" by Everett Baker, Shaun Bonefas, Brittany Kyer, Christian Morneau, Yan Wang and Shawn Burdette

The final findings were that the machine included in the Felicity tech tree, which is used for this process, would require 4.32MJ of energy and 5h of time to process one 100-kilogram batch of asteroid regolith into mixed metal alloy. The energy cost of the alloy was found to be 183.258 kJ/kg, and a single machine allows for production rate of 4.7146558 kg/h.

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1 Basic idea of the process

Finely powdered asteroid regolith is added to an aqueous solution of sodium hydroxide, at concentrations of 50% to 60% by mass. Current is passed through the solution at 1.7 volts to reduce the metal oxides into metals, and deposit them on the cathode. This takes about 5 hours to complete.

Once the process is done, the cathode is removed with the metals crystallized onto it, which are removed for further processing.

Only a select group of elements whose oxides break down at or below the voltage used can be produced this way, so other elements will remain in oxidized state and have to be extracted in some other way.

What I actually sourced from the study:

- The material to extract metal from needs to be powdered - that is the form in which material was processed in the study.
- A solution which is 50% to 60% sodium hydroxide by mass with water.
- A current is passed through the solution at 1.7 volts.
- 5 hour time to completion - that is how long a successful sample took to process in the study.
- The exact list of elements which can be extracted using this method - see figure 2 in the original study (on page 9).

Parts of this which were just my own guesses:

- The extracted metals will crystallize onto the cathode. I've seen this process many times whenever passing current through a solution results in formation of solid material, but that's about all that my reasoning for stating that this would happen is based on.
- The crystallized metals will deposit as a mixed alloy rather than separate layers. This is a conservative assumption as on the contrary assuming that the metals will deposit in layers at this stage would trivialize their later separation in a way that gives overly optimistic energy costs.

2 List of elements extracted via LTE

The listed elements are those which both can be extracted using Low Temperature Electrolysis and are present in asteroid regolith.

For reasons I'm about to explain, "asteroid regolith" and "meteor" had been used interchangeably here despite being obviously different, as that is an equivalence that my choice of my sources forced me to assume.

[Ptable](#) is an online interactive table of elements, which among other functions, provides the exact concentration of the chosen element as a percentage of mass in the whole universe, the solar system, meteor material, earth's crust, the ocean, and the human body. I chose the concentration in meteor material as likely the closest available approximation to asteroid material. Data was collected in 20.03.2025 and I had not re-checked it for any changes that might have happened since.

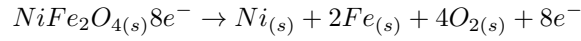
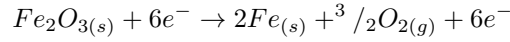
Atomic number	Symbol	Name	Percentage	Atomic mass	mass in 100kg
8	O	Oxygen	40	15.999	40kg
26	Fe	Iron	22	55.845	22kg
27	Co	Cobalt	0.059	58.93319	59g
28	Ni	Nickel	1.3	58.6934	1.3kg
29	Cu	Copper	0.011	63.546	11g
30	Zn	Zinc	0.018	65.38	18g
31	Ga	Gallium	0.00076	69.723	760mg
32	Ge	Germanium	0.0021	72.63	2.1g
33	As	Arsenic	0.00018	74.92160	180mg
34	Se	Selenium	0.0013	78.971	1.3g
42	Mo	Molybdenum	0.00012	95.95	120mg
44	Ru	Ruthenium	0.000081	101.07	81mg
45	Rh	Rhodium	0.000018	102.9055	16mg
46	Pd	Palladium	0.000066	106.42	66mg
47	Ag	Silver	0.000014	107.8682	14mg
48	Cd	Cadmium	0.000044	112.414	44mg
49	In	Indium	0.0000044	114.818	4.4mg
50	Sn	Tin	0.00012	118.710	120mg
51	Sb	Antimony	0.000012	121.760	12mg
52	Te	Tellurium	0.00021	127.60	210mg
74	W	Tungsten	0.000012	183.84	12mg
75	Re	Rhenium	0.000000049	186.207	4.9mg
76	Os	Osmium	0.000066	190.23	66mg
77	Ir	Iridium	0.000054	192.217	54mg
78	Pt	Platinum	0.000098	195.084	98mg
79	Au	Gold	0.000017	196.9666	17mg
80	Hg	Mercury	0.000025	200.59	25mg
81	Tl	Thallium	0.0000078	204.38	7.8mg
82	Pb	Lead	0.00014	207.2	140mg
83	Bi	Bismuth	0.0000069	208.9804	6.9mg

The expected yield from processing 100kg of dry asteroid regolith is **23.573279kg** of metal and 40kg of oxygen.

There will remain a sludge including 36.426721 kg of the remaining elements which could not be brought to their elemental, still oxidized and mixed with the sodium hydroxide solution whose mass had not been counted. This byproduct can be simply washed to recover and reuse the sodium hydroxide.

3 Energy and power use

The study presents 2 chemical processes that were used to determine how long to run the electrolysis process for:



My simplified interpretation here is that for every atom of iron to be split out of the iron oxide, 3 electrons pass from anode to cathode. In the second reaction, for every 1 atom of nickel and 2 atoms of iron, 8 electrons pass, which if we keep the assumption of 3 electrons per atom of iron means that 2 of these 8 electrons were used to free the nickel.

Since the voltage difference between anode and cathode is 1.7 volts, every electron carries 1.7 electron-volts. This can be used to estimate the energy needed to reduce the whole batch, if the total number of atoms is known.

For a real proper estimate I would have to know the expected number of electrons required to free the metal atom from the oxide compound for each of the 30 elements that are being extracted, which would require knowing what these oxides look like. For now such a task surpasses the limits of my dedication, so I will use the two provided formulas as an estimate, which is at the very least supported by the fact that iron and nickel are the two biggest mass fractions in the extracted metals.

22kg of iron and 1.3kg of nickel are extracted from electrolysis of 100kg of material.
Iron has a molar mass of 0.0558kg/mol, and Nickel has a molar mass of 0.05869kg/mol.

$$\frac{22kg}{0.0558kg/mol} = 394mol \quad \frac{1.3kg}{0.05869kg/mol} = 22.149mol$$

(I will likely eventually update the table in previous section with molar amounts calculated like this)

3 electrons are needed per iron atom, and 2 electrons per nickel atom:

$$394mol(Fe) \cdot \frac{3(e^-)}{1(Fe)} = 1182mol(e^-) \quad 22.149mol(Ni) \cdot \frac{2(e^-)}{1(Ni)} = 44.298mol(e^-)$$

In total, 1226.298 moles of electrons have to be transferred. 1 mole of electrons means $6.02214076 \times 10^{23}$ electrons in accordance with the avogadro constant. This means the total energy spent is:

$$1226.298mol \cdot \frac{6.02214076E + 23 \cdot 1.7eV}{mol} = 1.023764E + 24eV = 164025J$$

Now, that only accounts for the combined 23.3kg of iron and nickel, whereas the total mass of extracted metal is a bit over 23.57kg, and also assumes 100% efficiency. Majority of the remaining elements are heavier, and as such their molar fraction would be even smaller than their mass fraction, but as I had not done the calculations behind that, the mass fraction will do as a conservative estimate.

Additionally, I will assume a 50% efficiency, which feels reasonable for electrolysis, as the electrolysis of water typically has an efficiency well above that, though admittedly it is far from being the same thing.

$$164025J \cdot \frac{23.573279kg}{23.3kg} \cdot \frac{1}{0.5} = 331898J$$

So counting only electrolysis the energy cost is 331.898 kJ of electrical power for every 100kg batch of regolith.

In the study, successful electrolysis of a sample required 5 hours, which would mean that the electrolysis pulls just 18.44 watts, though small as that seems, at 1.7 volts that corresponds to nearly 10.85 amperes.

The solution needs to be hot, around 100°C, so it would use additional power just to stay hot. This can be mitigated using a vacuum insulation around the electrolysis cauldron, which would have the additional benefit of allowing the heat leakage to be simply estimated using the stefan-boltzmann law.

$$\sigma \cdot \epsilon \cdot A \cdot T^4 = P$$

where:

- σ is the stefan-boltzmann constant ($\approx 0.00000005670374419W \cdot m^{-2} \cdot K^{-4}$)
- ϵ is the emissivity of material used
- A is the surface area of the hot object
- T is the temperature of the hot object
- P is the power of thermal radiation, ie. rate at which the object's heat is lost to glowing at various wavelengths of light due to its heat

For smooth steel the emissivity is about 0.1.

If the electrolysis chamber should hold 100kg of asteroid material then it will likely hold about 200kg of total mass, might be about 200L in size, and if it were a cube that would give it a surface area of about 2m².

Temperature should be around 100°C which is 373.15K.

$$\sigma \cdot 0.1 \cdot 2m^2 \cdot (373.15K)^4 = 219.87W$$

So the real power use adds up to a little under 240W, with only about 18.5W of that being the electrolysis. This means that better insulation could still greatly decrease energy costs. Scaling up the process could do so as well, since the heat loss depends on surface area, and a bigger container would have much bigger ratio of volume to surface area.

$$5h \cdot 240W = 4.32MJ$$

So in the end the cost of refining one 100kg batch of regolith is: **4.32MJ** of energy and **5h** of time.

From another perspective, the cost of the metal alloy obtained this way is roughly **183.258 kJ/kg**, and a single machine can allow a maximum production rate of **4.7146558 kg/h**:

$$\frac{4.32MJ}{23.573279kg} = 183.258kJ/kg \quad \frac{23.573279kg}{5h} = 4.7146558kg/h$$

An interesting bit of trivia I can share that I originally did a little napkin math only loosely based on my vague estimates of the amperage and time required, which yielded 140MJ per batch, which as it turns out, overestimated the energy cost by a factor of 32.4 times.